STAT 331 Applied Linear Models

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Instructor: Kun Liang

LaTeX : Xing Liu

Contents

1 Simple Linear Regression (SLR)

Lecture 2, 2025/09/09

1.1 Introduction

Definition 1.1 (**SLR Population Model**).

Population model.

$$\text{Response: } y = \beta_0 + \beta_1 x + \epsilon,$$

where β_0, β_1 are regression coefficients, x is the predictor, and ϵ is the random error.

Definition 1.2 (**SLR Sample Model**).

Sample: n pairs of observations (x_i, y_i), $i = 1, 2, \dots, n$.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where β_0 is the intercept and β_1 is the slope.

	Fixed/Random	Known/Unknown
x_i's	Fixed	Known
β_i's	Fixed	Known
ϵ_i's	Random	Unknown
y_i's	Random	Known

SLR Assumptions

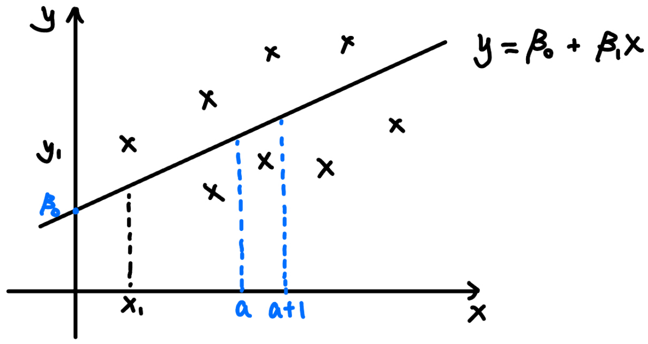
(1) $\mathbb{E}[\epsilon_i] = 0 \implies \mathbb{E}[y_i] = \beta_0 + \beta_1 x_i$.

(2) $\epsilon_1, \dots, \epsilon_n$ are statistically independent.

(3) $\operatorname{Var}(\epsilon_i) = \sigma^2 \implies \operatorname{Var}(y_i) = \sigma^2$.

(4) ϵ_i is normally distributed, i.e. $\epsilon_i \sim N(0, \sigma^2) \implies y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.

Note. The first three assumptions are called the **Gauss-Markov assumptions**.



The slope β_1 is often of primary interest. We have the following interpretations of β_1.

(1) $\beta_1 = \mathbb{E}[y \mid x = a+1] - \mathbb{E}[y \mid x = a]$.

(2) If $\beta_1 = 0$, then $\mathbb{E}[y \mid x] = \beta_0$, i.e. y does not depend on x.

1.2 Least Squares Estimation

Task: Given the sample observation (x_i, y_i), $i = 1, 2, \ldots, n$, we want to find (β_0, β_1) such that

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

is minimized.

To minimize the objective function $S(\beta_0, \beta_1)$, we set the partial derivatives to zero:

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0, \tag{1.1}$$

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) x_i = 0. \tag{1.2}$$

Then, to solve for β_0, β_1,

(1.1) $\implies n\beta_0 = \sum_{i=1}^{n} y_i - \beta_1 \sum_{i=1}^{n} x_i$, so

$$\beta_0 = \bar{y} - \beta_1 \bar{x}, \text{ where } \bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i \text{ and } \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

(2.1) $\implies \sum_{i=1}^n x_iy_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$. Substitute β_0 into this, we have

$$\sum_{i=1}^n x_iy_i - (\bar{y} - \beta_1\bar{x})\sum_{i=1}^n x_i - \beta_1\sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n x_iy_i - \bar{y}\sum_{i=1}^n x_i + \beta_1\bar{x}\sum_{i=1}^n x_i - \beta_1\sum_{i=1}^n x_i^2 = 0$$

$$\beta_1 = \frac{\sum_{i=1}^n x_iy_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}},$$

Lecture 3, 2025/09/11

Recall from STAT 231:

$$\text{Sample variance: } S_{xx} = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})^2,$$

$$\text{Sample covariance: } S_{xy} = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

Then,

$$\beta_1 = \frac{S_{xy}}{S_{xx}}.$$

Definition 1.3 (**Pearson Correlation Coefficient**).

The **Pearson correlation coefficient** between x and y is

$$\rho_{xy} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

Definition 1.4 (**Sample Correlation Coefficient**).

The **sample correlation coefficient** between x and y is

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}.$$

Then, we have $\beta_1 = r_{xy}\frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}}$.

Thus, the lease squares estimates (LSEs) are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Proposition 1.1 (Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$).

(1) The lease squares estimates (LSEs) are unbised, i.e. $\mathbb{E}\left[\hat{\beta}_0\right] = \beta_0, \mathbb{E}\left[\hat{\beta}_1\right] = \beta_1$.

(2) $\operatorname{Var}(\beta_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$, $\operatorname{Var}(\beta_1) = \frac{\sigma^2}{S_{xx}}$.

(3) $\operatorname{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) = -\frac{\sigma^2 \bar{x}}{S_{xx}}$.

Proof.

(1) We have

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}_1\right] &= \mathbb{E}\left[\frac{S_{xy}}{S_{xx}}\right] = \mathbb{E}\left[\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}\right] \\
&= \mathbb{E}\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right]
\end{aligned}$$

Note that $\hat{\beta}_1 = \displaystyle\sum_{i=1}^n c_i y_i$, where $c_i = \dfrac{x_i - \bar{x}}{S_{xx}}$.

Remark (**Properties of c_i's**).

(i) $\sum_{i=1}^n c_i = 0$.

(ii) $\sum_{i=1}^n c_i x_i = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i = \frac{1}{S_{xx}} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2\right) = 1$.

(iii) $\sum_{i=1}^n c_i^2 = \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{S_{xx}}$.

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}_1\right] &= \mathbb{E}\left[\sum_{i=1}^n c_i y_i\right] = \sum_{i=1}^n c_i \mathbb{E}\left[y_i\right] \\
&= \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i = \beta_1.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}_0\right] = \mathbb{E}\left[\bar{y} - \hat{\beta}_1 \bar{x}\right] &= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[y_i\right] - \bar{x}\mathbb{E}\left[\hat{\beta}_1\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}(\beta_0 + \beta_1 x_i) - \bar{x}\beta_1 \\
&= \beta_0 + \beta_1 \bar{x} - \bar{x}\beta_1 = \beta_0.
\end{aligned}$$

(2) We have

$$\begin{aligned}
\operatorname{Var}\left(\hat{\beta}_1\right) &= \operatorname{Var}\left(\sum_{i=1}^{n} c_i y_i\right) = \sum_{i=1}^{n} c_i^2 \operatorname{Var}\left(y_i\right) \\
&= \sigma^2 \sum_{i=1}^{n} c_i^2 = \frac{\sigma^2}{S_{xx}}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\operatorname{Var}\left(\hat{\beta}_0\right) &= \operatorname{Var}\left(\bar{y} - \hat{\beta}_1 \bar{x}\right) = \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} y_i - \bar{x}\sum_{i=1}^{n} c_i y_i\right) \\
&= \operatorname{Var}\left(\sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) y_i\right) = \sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right)^2 \operatorname{Var}\left(y_i\right) \\
&= \sigma^2 \sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right)^2 = \sigma^2\left(\sum_{i=1}^{n}\frac{1}{n^2} - 2\frac{\bar{x}}{n}\sum_{i=1}^{n} c_i + \bar{x}^2\sum_{i=1}^{n} c_i^2\right) \\
&= \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right).
\end{aligned}$$

(3) We have

$$\begin{aligned}
\operatorname{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) &= \operatorname{Cov}\left(\bar{y} - \hat{\beta}_1\bar{x}, \hat{\beta}_1\right) \\
&= \operatorname{Cov}\left(\sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) y_i, \sum_{i=1}^{n} c_i y_i\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) c_j \operatorname{Cov}\left(y_i, y_j\right)
\end{aligned}$$

Note that $\mathrm{Cov}\left(y_i, y_j\right) = \begin{cases} \sigma^2, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$ Then,

$$\begin{aligned}
\mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) &= \sigma^2 \sum_{i=1}^{n} \left(\frac{1}{n} - c_i \bar{x}\right) c_i \\
&= \sigma^2 \left(\frac{1}{n} \sum_{i=1}^{n} c_i - \bar{x} \sum_{i=1}^{n} c_i^2\right) = -\frac{\sigma^2 \bar{x}}{S_{xx}}.
\end{aligned}$$

□

Definition 1.5 (Residual).

The **residual** for the i-th observation is the difference between the observed and the fitted value:

$$r_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

Under the least squares fit, we have the following.

Proposition 1.2 (Properties of Residuals).

(1) $\sum_{i=1}^{n} r_i = 0$.

(2) $\sum_{i=1}^{n} r_i x_i = 0$.

(3) $\sum_{i=1}^{n} r_i \hat{y}_i = 0$, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

Lecture 4, 2025/09/16

To find the estimator of σ^2, note that

$$\begin{aligned}
\epsilon_i &= y_i - (\beta_0 + \beta_1 x_i) \\
r_i &= y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i).
\end{aligned}$$

If ϵ_i's are known, we can estimate σ^2 by $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} \epsilon_i^2$ because $\epsilon_i \sim N(0, \sigma^2)$ and

$$\mathbb{E}\left[\hat{\sigma}^2\right] = \mathbb{E}\left[\epsilon_i\right]^2 + \mathrm{Var}\left(\epsilon_i\right) = \sigma^2.$$

However $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} r_i^2\right] \neq \sigma^2$. If we define

$$s^2 = \frac{1}{n-2}\sum_{i=1}^{n} r_i^2,$$

then, $\mathbb{E}\left[s^2\right] = \sigma^2$.

Note. The degree of freedom is $n-2$ because we have estimated two parameters, β_0 and β_1.

Example (**Degree of Freedom Example**).
If $y_i \sim N(\mu, \sigma^2)$, then the sample variance estimator is

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \hat{\mu})^2,$$

where the degree of freedom is $n-1$ because we have estimated one parameter. Note that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$.

1.3 SLR: Inference

Proposition 1.3. Under the assumptions that ϵ_i's are independent and normally distributed,

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right).$$

Remark. If ϵ^2 is known, then

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0,1).$$

In most practice, σ^2 is unknown, and we replace σ^2 by s^2:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2/S_{xx}}} \sim t_{n-2}.$$

Proposition 1.4 (**Confidence Interval for β_1**).
A $100(1-\alpha)\%$ confidence interval (CI) for β_1 is

$$[\hat{\beta}_1 - t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right), \hat{\beta}_1 + t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right)],$$

or equivalently

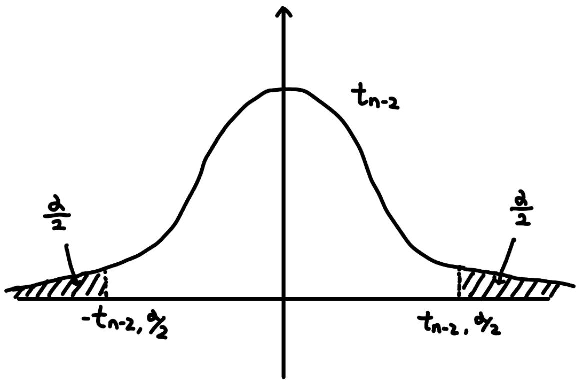
$$\hat{\beta}_1 \pm t_{n-2,\alpha/2}\, \mathrm{SE}\left(\hat{\beta}_1\right), \text{ where } \mathrm{SE}\left(\hat{\beta}_1\right) = \frac{s}{\sqrt{S_{xx}}}.$$

Proof. Note that

$$\mathbb{P}\left(-t_{n-2,\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\mathrm{SE}\left(\hat{\beta}_1\right)} \leq t_{n-2,\alpha/2}\right) = 1 - \alpha$$

$$\mathbb{P}\left(\hat{\beta}_1 - t_{n-2,\alpha/2}\, \mathrm{SE}\left(\hat{\beta}_1\right) \leq \beta_1 \leq \hat{\beta}_1 + t_{n-2,\alpha/2}\, \mathrm{SE}\left(\hat{\beta}_1\right)\right) = 1 - \alpha.$$

□



Hypothesis Testing for β_1

We test

$$H_0 : \beta_1 = \beta_1^* \text{ vs. } H_a : \beta_1 \neq \beta_1^*.$$

Under H_0, the test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\mathrm{SE}\left(\hat{\beta}_1\right)} \sim t_{n-2}.$$

Reject H_0 at significance level α if $|t| > t_{n-2,\alpha/2}$. Alternatively, compute the p-value:

$$p = 2\mathbb{P}(T > |t|) \quad \text{where } T \sim t_{n-2}$$

and reject H_0 if $p < \alpha$. Typically, we are interested in testing $H_0 : \beta_1 = 0$.

Note. $p = \mathbb{P}(|T| \geq |t|) = 2\mathbb{P}(T \geq |t|)$ because t-distribution is symmetric about 0.

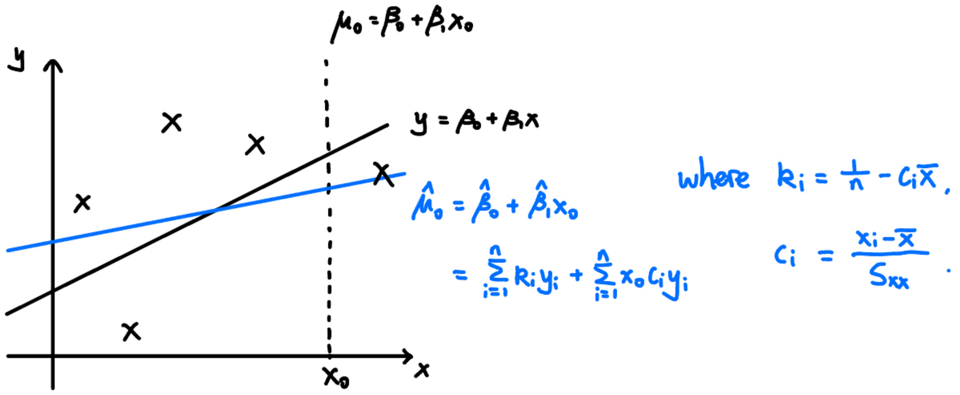
Inference for Mean Response

For the predictor value x_0, the mean response is $\mu_0 = \beta_0 + \beta_1 x_0$. We estimate μ_0 by

$$\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

We can use $\hat{\mu}_0 = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_0 = \bar{y} + \hat{\beta}_1 (x_0 - \bar{x})$ to show that

$$\hat{\mu}_0 = \sum_i d_i y_i, \text{ where } d_i = \frac{1}{n} + \frac{(x_i - \bar{x})(x_0 - \bar{x})}{S_{xx}}.$$



Note. The black line is the true regression line and the blue line is the fitted line based on estimates.

Proposition 1.5 (**Expectation & Variance of** $\hat{\mu}_0$).

For $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$,

$$\mathbb{E}[\hat{\mu}_0] = \mu_0, \quad \operatorname{Var}(\hat{\mu}_0) = \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right] \sigma^2.$$

Question: What is your best guess of y given $x = x_p$?

Answer: $\hat{y}_p$, where $\hat{y}_p = \hat{\beta}_0 + \hat{\beta}_1 x_p$.

Note that

$$\mathbb{E}[y_p - \hat{y}_p] = 0, \quad \operatorname{Var}(y_p - \hat{y}_p) = \left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right] \sigma^2.$$

Since σ^2 is unknown, we use s^2 and get

$$\frac{y_p - \hat{y}_p}{\operatorname{SE}(y_p - \hat{y}_p)} \sim t_{n-2}, \text{ where } \operatorname{SE}(y_p - \hat{y}_p) = s\sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}.$$

Note. $y_p = \beta_0 + \beta_1 x_p + \epsilon_p$ is a new observation at $x = x_p$.

Proposition 1.6 (**Prediction Interval for y_p**).
A $100(1-\alpha)\%$ prediction interval (PI) for y_p is

$$\hat{y}_p \pm t_{n-2,\alpha/2}\,\mathrm{SE}\left(y_p - \hat{y}_p\right), \text{ where } \mathrm{SE}\left(y_p - \hat{y}_p\right) = s\sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}.$$

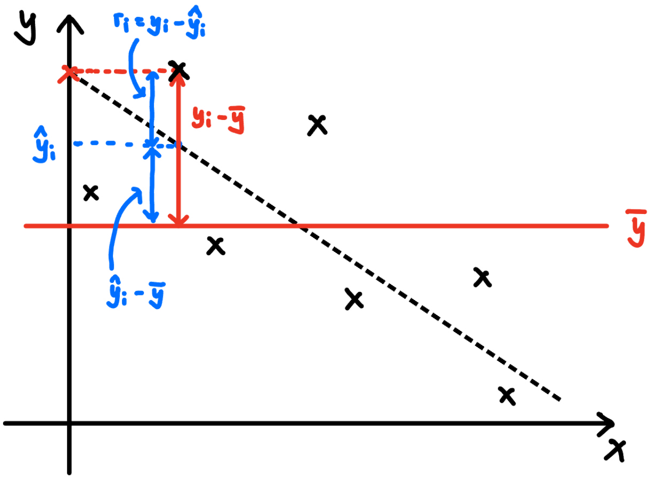
Lecture 5, 2025/09/18

1.4 SLR: ANOVA and Cochran's Theorem

Analysis of Variance (ANOVA): How well does our regression model fit the response variable?

Total variation in y_i's is measured by the total sum of squares:

$$\mathrm{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2.$$



Proposition 1.7 (**ANOVA Decomposition**).
We have the following decomposition.

$$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE},$$

where

$$\text{SSR} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \text{ is the regression sum of squares,}$$
$$\text{SSE} = \sum_{i=1}^{n} r_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \text{ is the residual sum of squares.}$$

Proof. Notice that $y_i - \bar{y} = y_i - \hat{y}_i + \hat{y}_i - \bar{y}$. Now, we have

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^{n} (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{n} (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\
&= \text{SSE} + \text{SSR} + 2\sum_{i=1}^{n} r_i(\hat{y}_i - \bar{y}) \\
&= \text{SSE} + \text{SSR}\,.
\end{aligned}$$

□

Note. SSR is the variability explained by the model; SSE is the variability not explained by the model.

ANOVA for Testing $H_0 : \beta_1 = 0$

If H_0 is true, $y_i = \beta_0 + \epsilon_i$. SSR should be "small" relatively to SSE.

Proposition 1.8. Under $H_0 : \beta_1 = 0$, we have the following.

(1) $\frac{\text{SSR}}{\sigma^2} \sim \chi_1^2$.

(2) $\frac{\text{SST}}{\sigma^2} \sim \chi_{n-1}^2$.

(3) $\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-2}^2$, and SSE is independent of SSR.

Proof.

(1) We have

$$\begin{aligned}
\text{SSR} &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2 \\
&= \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{xx}.
\end{aligned}$$

Recall that $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$. Then, we have $\left(\frac{\hat{\beta}_1-\beta_1}{\sigma/\sqrt{S_{xx}}}\right)^2 \sim \chi_1^2$. Under $H_0 : \beta_1 = 0$, we have

$$\frac{\hat{\beta}_1^2 S_{xx}}{\sigma^2} \sim \chi_1^2 \implies \frac{\text{SSR}}{\sigma^2} \sim \chi_1^2.$$

(2) Under $H_0 : \beta_1 = 0$, we have $y_1, \dots, y_n \overset{\text{iid}}{\sim} N(\beta_0, \sigma^2)$. That is, $\frac{y_i-\beta_0}{\sigma} \sim N(0,1)$. Thus,

$$\sum_{i=1}^n \left(\frac{y_i - \beta_0}{\sigma}\right)^2 \sim \chi_n^2.$$

Furthermore, $\left(\frac{\bar{y}-\beta_0}{\sigma/\sqrt{n}}\right)^2 \sim \chi_1^2$ because $\bar{y} \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$. Now,

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\
&= \sum_{i=1}^n \left[(y_i - \beta_0) - (\bar{y} - \beta_0)\right]^2 \\
&= \sum_{i=1}^n (y_i - \beta_0)^2 - n(\bar{y} - \beta_0)^2.
\end{aligned}$$

Thus, we have

$$\frac{\sum_{i=1}^n (y_i - \beta_0)^2}{\sigma^2} = \frac{\text{SST}}{\sigma^2} + \left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}}\right)^2 \iff S = Q_1 + Q_2,$$

where $S \sim \chi_n^2$. For $Q_1 = \frac{\text{SST}}{\sigma^2}$, there are only $n-1$ degrees of freedom because $\sum_{i=1}^n (y_i - \bar{y}) = 0$. For $Q_2 = \left(\frac{\bar{y}-\beta_0}{\sigma/\sqrt{n}}\right)^2$, we only have $\bar{y}$ and so $d_2 = 1$. By Cochran's theorem, Q_1 and Q_2 are independent, and $\frac{\text{SST}}{\sigma^2} \sim \chi_{n-1}^2$.

(3) We have

$$\begin{aligned}
\frac{\text{SST}}{\sigma^2} &= \frac{\text{SSE}}{\sigma^2} + \frac{\text{SSR}}{\sigma^2} \\
\underbrace{S}_{\sim \chi_{n-1}^2} &= Q_1 + \underbrace{Q_2}_{\sim \chi_1^2}.
\end{aligned}$$

The degrees of freedom of $\text{SSE} = \sum_{i=1}^n r_i^2$ is $n-2$ because $\sum_{i=1}^n r_i = 0$ and $\sum_{i=1}^n r_i x_i = 0$. By Cochran's theorem, $\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-2}^2$ and SSE is independent of SSR.

$\square$

Theorem 1.9 (**Cochran's Theorem**).
Suppose that $U_i \overset{\text{iid}}{\sim} N(0,1)$ for $i = 1, 2, \ldots, n$, and

$$\sum_{i=1}^{n} U_i^2 = Q_1 + Q_2.$$

Let d_1 and d_2 be the degrees of freedom of Q_1 and Q_2, i.e. the numbers of linearly independent linear combinations of the U_i's in Q_1 and Q_2. If $d_1 + d_2 = n$, then Q_1 and Q_2 are independent, and furthermore,

$$Q_1 \sim \chi^2_{d_1}, \quad Q_2 \sim \chi^2_{d_2}.$$

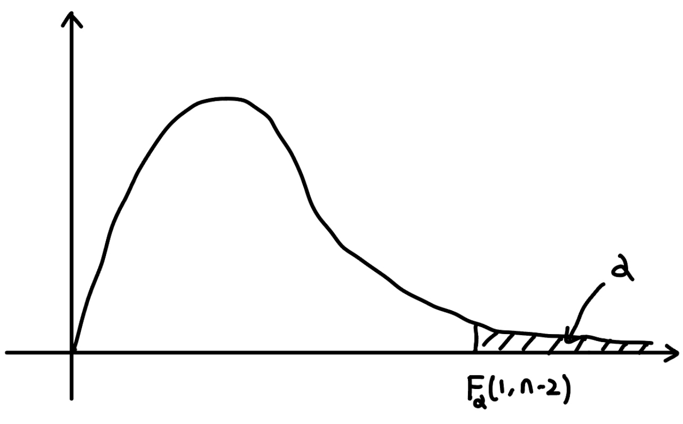
Proposition 1.8 above allows us to define the following F-statistic (with $\beta_1 = 0$ under H_0):

$$F = \frac{\frac{\text{SSR}}{\sigma^2}/1}{\frac{\text{SSE}}{\sigma^2}/(n-2)} = \frac{\text{SSR}}{\text{SSE}\,/(n-2)} = \frac{\text{MSR}}{\text{MSE}} \sim F(1, n-2).$$

Note that MSR is the mean squares of regression and MSE is the mean squares of error. To test $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$, we reject H_0 at the level α if

$$F > F_\alpha(1, n-2), \text{ where } F_\alpha(1, n-2) \text{ is the } 100(1-\alpha)\%\text{-tile of } F(1, n-2).$$

The F-distribution looks like this:



To test $H_0 : \beta_1 = 0$, we use the F-statistic:

$$F = \frac{\text{SSR}}{\text{SSE}/(n-2)} = \frac{\text{MSR}}{\text{MSE}} \sim F(1, n-2).$$

Recall that

$$\frac{\hat{\beta}_1 - \beta_1^*}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

and we can use this to test $H_0 L \beta_1 = \beta_1^*$. In SLR, testing $H_0 : \beta_1 = 0$ using t-test and F-test are equivalent. If $X \sim t_{n-2}$, then $X^2 \sim F(1, n-2)$.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Regression	SSR	1	$\text{MSR} = \frac{\text{SSR}}{1}$	$\frac{\text{MSR}}{\text{MSE}}$
Residual	SSE	$n-2$	$\text{MSE} = \frac{\text{SSE}}{n-2}$	
Total	SST	$n-1$		

Definition 1.6 (**Coefficient of Determination**).
The **coefficient of determination** is defined as

$$R^2 = \frac{\text{SSR}}{\text{SST}}.$$

It measures the proportion of total variation in the response variable y around the mean that is explained by the regression model.

$0 \le R^2 \le 1$, $R^2 = 0$ means none of the variability in y is explained by x. In SLR, we have

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\hat{\beta}_1^2 S_{xx}}{S_{yy}} = \frac{\left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{yy} S_{xx}}.$$

Recall that the sample correlation coefficient is

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

and thus $R^2 = r^2$.

2 Multiple Linear Regression (MLR)

2.1 Linear Algebra Review and MVN Distribution

Definition 2.1 (Random Vectors).

Let $\mathbf{Y} = (y_1, \dots, y_n)^\top$ be a vector of random variables. Then, $\mathbf{Y}$ is called a **random vector**. Note that

$$\mathbb{E}[y_i] = \mu_i, \quad \operatorname{Var}(y_i) = \sigma_i^2, \quad \operatorname{Cov}(y_i, y_j) = \sigma_{ij}.$$

Also,

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}, \quad \operatorname{Var}(\mathbf{Y}) = \operatorname{Cov}(\mathbf{Y}, \mathbf{Y}) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix}.$$

Remark. Note that

$$\begin{aligned}\operatorname{Var}(\mathbf{Y}) &= \mathbb{E}\left[(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^\top\right] \\ &= \begin{pmatrix} \operatorname{Var}(y_1) & \operatorname{Cov}(y_1, y_2) & \cdots & \operatorname{Cov}(y_1, y_n) \\ \operatorname{Cov}(y_2, y_1) & \operatorname{Var}(y_2) & \cdots & \operatorname{Cov}(y_2, y_n) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{Cov}(y_n, y_1) & \operatorname{Cov}(y_n, y_2) & \cdots & \operatorname{Var}(y_n) \end{pmatrix}\end{aligned}$$

Example. If $y_1, \dots, y_n$ are independent, $\operatorname{Cov}(y_i, y_j) = 0$ $\forall i \neq j$. Furthermore, $\operatorname{Var}(y_i) = \sigma^2$ $\forall i$. This implies that

$$\operatorname{Var}(\mathbf{Y}) = \sigma^2 I, \qquad \text{where } I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \text{ is the identity matrix.}$$

Proposition 2.1 (**Basic Properties**).

Suppose $A = (a_{ij})_{m\times n}$, $\mathbf{b} = (b_1, \dots, b_m)^\top$, $\mathbf{c} = (c_1, \dots, c_m)^\top$ are matrix and vectors of constants. Then,

(1) $\mathbb{E}\left[A\mathbf{Y} + \mathbf{b}\right] = A\mathbb{E}\left[\mathbf{Y}\right] + \mathbf{b}$.

(2) $\operatorname{Var}(\mathbf{Y} + \mathbf{c}) \operatorname{Var}(\mathbf{Y})$.

(3) $\operatorname{Var}(A\mathbf{Y}) = A \operatorname{Var}(\mathbf{Y}) A^\top$.

(4) $\operatorname{Var}(A\mathbf{Y} + \mathbf{b}) = A \operatorname{Var}(\mathbf{Y}) A^\top$.

Definition 2.2 (**Differentiation: Linear & Quadratic Forms**).

Scalar f with respect to the vector $\mathbf{Y}$:

(1) If $f = f(\mathbf{Y})$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = \begin{pmatrix} \frac{\mathrm{d}f}{\mathrm{d}y_1} \\ \vdots \\ \frac{\mathrm{d}f}{\mathrm{d}y_n} \end{pmatrix}.$$

(2) If $f = \mathbf{c}^\top\mathbf{Y} = \sum c_i y_i$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = \mathbf{c}.$$

(3) If $A = A^\top$ and $f = \mathbf{Y}^\top A\mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} y_i y_j$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = 2A\mathbf{Y}.$$

In this case, f is called a **quadratic form**.

Definition 2.3 (**Trace**).

The **trace** of a square matrix $A = (a_{ij})_{n\times n}$ is defined as

$$\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}.$$

Remark. We have $\operatorname{tr}(BC) = \operatorname{tr}(CB)$.

Definition 2.4 (**Rank**).

The **rank** of a matrix A is defined as

$$\operatorname{rank}(A) = \text{the maximum number of linearly independent columns (or rows) of } A.$$

Definition 2.5 (**Linearly Independence**).

Random vectors $\mathbf{Y}_1, \ldots, \mathbf{Y}_n$ are linearly independent $\iff$

$$c_1\mathbf{Y}_1 + \cdots + c_n\mathbf{Y}_n = 0 \implies c_1 = c_2 = \cdots = c_n = 0.$$

Definition 2.6 (**Orthogonality, Orthogonal Matrix**).

Random vectors $\mathbf{X}, \mathbf{Y}$ are **orthogonal** if $\mathbf{X}^\top\mathbf{Y} = 0$.

A square matrix A is an **orthogonal matrix** if $A^\top A = AA^\top = I$.

Definition 2.7 (**Eigenvalues, Eigenvectors**).

Let A be a square matrix. A scalar λ is an **eigenvalue** of A if there exists a non-zero vector $\mathbf{v}$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is called an **eigenvector** corresponding to the eigenvalue λ.

Definition 2.8 (**Idempotent Matrix**).

A square matrix A is **idempotent** if $A^2 = A$.

Proposition 2.2. Let A be an idempotent matrix.

(1) All eigenvalues of A are either 0 or 1.

(2) If A is symmetric (i.e. $A = A^\top$), then $\exists$ orthogonal P s.t.

$$A = P\Lambda P^\top, \quad \text{where } \Lambda = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0).$$

Then, $\operatorname{tr}(A) = \operatorname{rank}(A) = \#$ of eigenvalues equal to 1.

Proof.

(1) Let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{v}$. Then,

$$A^2\mathbf{v} = A(A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda A\mathbf{v} = \lambda^2\mathbf{v}.$$

But since A is idempotent, $A^2\mathbf{v} = A\mathbf{v} = \lambda\mathbf{v}$. Thus,

$$\lambda^2\mathbf{v} = \lambda\mathbf{v} \implies \lambda(\lambda - 1)\mathbf{v} = 0.$$

(2) For the $\mathrm{tr}(A) = \mathrm{rank}(A)$ part, note that

$$\mathrm{tr}(A) = \mathrm{tr}\left(P\Lambda P^\top\right) = \mathrm{tr}\left(\Lambda P^\top P\right) = \mathrm{tr}(\Lambda) = \mathrm{rank}(A).$$

$\square$

Definition 2.9 (**Multivariate Normal Distribution (MVN)**).

$\mathbf{Y} \sim \mathrm{MVN}(\mu, \Sigma)$ if $\mathbf{Y}$ has the following probability density function (PDF):

$$f(\mathbf{y}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mu)^\top\Sigma^{-1}(\mathbf{y} - \mu)\right).$$

Also, $\mathbb{E}\left[\mathbf{Y}\right] = \mu$ and $\mathrm{Var}\left(\mathbf{Y}\right) = \Sigma$.

Proposition 2.3 (**Properties of MVN**).

Suppose $\mathbf{Y}$ is a random vector and A, B are matrices. Then,

(1) $y_1, \ldots, y_n$ are independent $\iff$ Σ is diagonal.

(2) Marginal normality: $y_i \sim \mathrm{N}(\mu_i, \sigma_{ii})$.

(3) If $\mathbf{Y} \sim \mathrm{MVN}(\mu, \Sigma)$ and $Z = A\mathbf{Y}$, then

$$Z \sim \mathrm{MVN}(A\mu, A\Sigma A^\top).$$

(4) If $\mathbf{Y} \sim \mathrm{MVN}(0, \sigma^2 I)$, then

$$\frac{1}{\sigma^2}\mathbf{Y}^\top\mathbf{Y} \sim \chi^2_n.$$

(5) If $U = A\mathbf{Y}$ and $W = B\mathbf{Y}$, then

$$\mathrm{Cov}\,(U, W) = A\Sigma B^\top.$$

Then, $U \perp\!\!\!\perp W \iff A\Sigma B^\top = 0$.

Remark. For (5), note that

$$\mathrm{Cov}\,(U, W) = \mathbb{E}\left[(U - \mathbb{E}\,[U])(W - \mathbb{E}\,[W])^\top\right] = A\mathbb{E}\left[(\mathbf{Y} - \mathbb{E}\,[\mathbf{Y}])(\mathbf{Y} - \mathbb{E}\,[\mathbf{Y}])^\top\right] B^\top = A\Sigma B^\top.$$